Bhisham Dev Verma — Curriculum Vitae

Research Interests

Dimension Reduction/Sketching Algorithms, Randomized Numerical Linear Algebra, Scalable Algorithms for Data Science, Parallel Computing, Soft Computing and Machine Learning.

Work Experience

Postdoctoral Research Associate — *August 2024 - present*
Wake Forest University,
Winston-Salem, North Carolina, United States.
Advisor: Dr. Grey Ballard

Education

Doctor of Philosophy (Ph.D.) — *February 2019 - June 2024*
Indian Institute of Technology Mandi,
Himachal Pradesh, India.
Thesis: Improving Sketching Algorithms via Statistical Variance Reduction Techniques
Advisors: Prof. Manoj Thakur and Dr. Rameshwar Pratap

Master of Science in Applied Mathematics — *August 2016 - June 2018*
Indian Institute of Technology Mandi,
Himachal Pradesh, India.
Thesis: Practical Portfolio Optimization using Heuristic Optimization Techniques
Advisor: Prof. Manoj Thakur

Bachelor of Science (B.Sc.) — *August 2012 - March 2015*
Himachal Pradesh University, Shimla, India.

Journal Publications

1. *Stochastic trace and diagonal estimator for tensors.* **Bhisham Dev Verma**, Rameshwar Pratap, and Keegan Kang. *To appear, Theoretical Computer Science (TCS) journal, 2026.*
2. ****Adaptive randomized tensor train rounding using Khatri-Rao products.* Hussam Al Daas, Grey Ballard, Laura Grigori, Mariana Martinez Aguilar, Arvind K. Saibaba, and **Bhisham Dev Verma**. *To appear, SIAM Journal on Scientific Computing (SISC), 2026.*
3. *Efficient CP rounding using alternating least squares with QR decomposition.* Alex Zhang, **Bhisham Dev Verma**, Jan Van Lent, and Grey Ballard. *SIAM Journal on Matrix Analysis and Applications (SIMAX) 47.2 (2026): 544-563*
4. *Faster and space efficient indexing for locality sensitive hashing.* **Bhisham Dev Verma** and Rameshwar Pratap. *Accepted in the Theoretical Computer Science (TCS) journal (2025): 115479.*
5. *Credibilistic skewness of LR power fuzzy numbers with applications in portfolio selection.* Pawan Kumar Mandal, **Bhisham Dev Verma**, Manoj Thakur and Garima Mittal. *Applied Intelligence 55.12 (2025): 839.*
6. *Improving LSH via tensorized random projection.* **Bhisham Dev Verma** and Rameshwar Pratap. *Acta Informatica 62.1 (2025): 1-35.*
7. *An ant interaction scheme-based wrapper strategy for hyperspectral band selection.* Kamal Deep, **Bhisham Dev Verma**, and Manoj Thakur. *Infrared Physics & Technology 145 (2025): 105726.*
8. *Improving compressed matrix multiplication using control variate Method.* **Bhisham Dev Verma**, Punit Pankaj Dubey, Rameshwar Pratap and Manoj Thakur. *Information Processing Letters 187 (2025): 106517.*
9. *Sparsifying Count Sketch.* **Bhisham Dev Verma**, Rameshwar Pratap and Punit Pankaj Dubey. *Information Processing Letters 186 (2024): 106490.*

10. *Unbiased estimation of inner product via higher order count sketch.* **Bhisham Dev Verma**, Rameshwar Pratap and Manoj Thakur. *Information Processing Letters 183 (2023): 106407.*
11. *Variance reduction in feature hashing using MLE and control variate method.* **Bhisham Dev Verma**, Rameshwar Pratap and Manoj Thakur. *Machine Learning 111.7 (2022): 2631-2662.*
12. *Efficient binary embedding of categorical data using BinSketch.* **Bhisham Dev Verma**, Rameshwar Pratap and Debajyoti Bera. *Data Mining and Knowledge Discovery 36.2 (2022): 537-565.*
13. *Dimensionality reduction for categorical data.* Debajyoti Bera, Rameshwar Pratap, and **Bhisham Dev Verma**. *IEEE Transactions on Knowledge and Data Engineering 35.4 (2021): 3658-3671.*
14. *QUINT: Node embedding using network hashing.* Debajyoti Bera, Rameshwar Pratap, **Bhisham Dev Verma**, Biswadeep Sen and Tanmoy Chakraborty. *IEEE Transactions on Knowledge and Data Engineering 35.3 (2021): 2987-3000.*
15. *Parallel multi-agent real-coded genetic algorithm for large-scale black-box single-objective optimisation.* Andranik S. Akopov, Levon A. Beklaryan, Manoj Thakur, and **Bhisham Dev Verma**. *Knowledge-Based Systems 174 (2019): 103-122.*

** Author order is alphabetical by last name. I am a primary contributor and the corresponding author of this work.

Conference Publications

16. *It's all in the (exponential) family: An equivalence between maximum likelihood estimation and control variates for sketching algorithms.* Keegan Kang, Kerong Wang, Ding Zhang, Rameshwar Pratap, **Bhisham Dev Verma**, Benedict H. W. Wong. *Accepted to the 29th Annual Conference on Artificial Intelligence and Statistics (AISTAT 2026).*
17. *Improving Sign-Random-Projection via count sketch.* Punit Pankaj Dubey, **Bhisham Dev Verma**, Rameshwar Pratap, and Keegan Kang. *In the 38th Conference on Uncertainty in Artificial Intelligence (UAI), pages 599-609, 2022.*
18. *Improving Tug-of-War sketch using control variate method.* Rameshwar Pratap, **Bhisham Dev Verma**, and Raghav Kulkarni. *In the SIAM Conference on Applied and Computational Discrete Algorithms (ACDA21), pages 66-76, 2021.*

Works in preparation / under review

1. *Improved analysis of Khatri-Rao random projections and applications.* Arvind K. Saibaba, **Bhisham Dev Verma**, and Grey Ballard. *Revision submitted to the SIAM Journal on Matrix Analysis and Applications (SIMAX), March 2026.*
2. *** Mixed Precision Compression of Tucker Decomposition.* Grey Ballard, Theo Mary, and **Bhisham Dev Verma**. *Submitted to the ACM TOMS, June 2026.*

Teaching Experience

- Teaching Assistant at IIT Mandi in Applied Mathematical Programming (Aug-Dec 2023), Probability & Statistics (Feb-July 2023), Matrix Computations for Data Science (Feb-July 2022, Aug-Dec 2020), Scalable Data Science (Aug-Dec 2021), Statistical Foundations of Data Science (Feb-July 2021), Programming Practicum (Feb-July 2020), Data Science 1 (Aug-Dec 2019), Optimization Techniques (Feb-June 2019).

Talks

- *"Adaptive Randomized Tensor Train Rounding using Khatri-Rao Products"* at ILAS 2026, Virginia Tech, May 2026.
- *"Adaptive Randomized Algorithm for Rounding in the Tensor-Train Format"* at Sparsitute All Members Meeting, Berkeley, California, November 2025.
- *"Improving Tug-of-War sketch using Control-Variates method"* at SIAM Conference on Applied and Computational Discrete Algorithms (ACDA21), July 2021.

Workshop & Summer schools

- Attended Workshop on Algorithms under Uncertainty (16-17 Dec 2022) organised by the Department of Computer Science and Engineering, Indian Institute of Technology Madras, Chennai, India.
- Attended Research Week with Google organised by Google Research India, Feb 8-11, 2022.
- Attended Google Research India Graduate Symposium, April 7-9, 2021.
- Attended IFCAM Summer School Workshop 2019 on Mathematics For Data Science organised by IISc Banglore.
- Attended Summer Program in Mathematics-2017 organised by HRI Allahabad.

Honors and Award

- Awarded Best Teaching Assistant Award-2021 by the Indian Institute of Technology Mandi in recognition of contribution in the preparation of assignments and clearing doubts of the students in the Data Science 1 course.
- Awarded Outstanding Academic Achievement Award for academic excellence among the graduating batch of M.Sc. Applied Mathematics students in the year 2018.
- Qualified Graduate Aptitude Test in Engineering (GATE) 2018 in Mathematics.
- Awarded Merit cum Mean Scholarship by IIT Mandi for the period of one year, from August 2016 to July 2017.

Web Links

Homepage, Google Scholar